

ABL Customer Statement Intelligence Suite

PRODUCTION FUNCTIONAL

Final Technology Stack

Layer / Component	Selected Technology	Role in Final System
Frontend	React.js + Tailwind CSS	Unified conversational workspace and responsive UI.
Build Tool	Vite	Frontend development and optimized production build.
Frontend Hosting	Vercel	Production hosting for the React/Vite single-page application.
Backend	FastAPI	Typed REST APIs, validation, orchestration, and banking-intelligence services.
Backend Language	Python 3.12	Authoritative financial calculations and deterministic business rules.
Backend Hosting	Heroku	Production FastAPI deployment; current validated release is v17.
LLM Provider / Model	Groq openai/gpt-oss-20b	Grounded explanation and interpretation only; not the authority for financial arithmetic.
Embedding Runtime	FastEmbed / ONNX	Lower-memory semantic embedding path suitable for the production backend.
Embedding Model	all-MiniLM-L6-v2 384 dimensions	Semantic representation of public-policy questions and corpus entries.
Database	PostgreSQL	Policy corpus and embedding storage.
Vector Search	pgvector	Similarity search inside PostgreSQL without a separate vector database.
PostgreSQL Driver	psycopg	Python/PostgreSQL connectivity.
Version Control / CI	Git + GitHub Actions	Source control and automated validation.

156/156
backend tests passing

a04d7b1
production code commit

47d72d9
current main docs sync

Heroku v17
validated production release

Final architecture. A React/Vite frontend on Vercel communicates with a modular FastAPI backend on Heroku. A deterministic unified router selects statement intelligence, recurring-payment rules, or public-policy RAG. Python remains authoritative for financial calculations; Groq is used only for grounded language generation. Policy retrieval uses FastEmbed MiniLM embeddings with PostgreSQL + pgvector.

Primary conversational endpoint

POST /assistant/chat

Specialized endpoints remain available for backward compatibility.

Input hardening

CSV max 2 MiB · 5000 rows · question max 1000 chars · description max 500 chars · history max 8 messages.

Internship prototype — not an official Allied Bank customer website.
This independent student project is a synthetic-data demonstration created for educational and internship purposes. Do not upload real customer statements, credentials, or confidential information.
Prototype boundary: synthetic statement data and a limited corpus of public Allied Bank information only; no live core-banking integration or production customer authentication.